Zhenghao He

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RESEARCH INTERESTS

- Test-time reasoning, controllability, and inference-time scaling in foundation models
- Mechanistic interpretability and activation-level steering of LLMs
- Reliable reasoning systems across autoregressive and diffusion language models

EDUCATION

University of Virginia — **Charlottesville, United States**
Doctor of Philosophy in Computer Science (3.9 / 4.0 GPA) — *Aug 2024 – May 2029 (expected)*
- Advisor: Prof. Aidong Zhang

Tongji University — **Shanghai, China**
Bachelor of Engineering in Computer Science and Technology(88.50/100 GPA) — *Sept 2020 – July 2024*
- Relevant Coursework: Machine Learning, Deep Learning, Data Mining, Computer Vision, Algorithms.

ONGOING RESEARCH

Reasoning and Test-Time Inference in Diffusion Language Models
- Investigated reasoning dynamics and token commitment order in diffusion language models, identifying premature answer commitment as a failure mode in multi-step reasoning.
- Developed counterfactual masking and directional-dependency analyses to quantify information flow between supporting reasoning and conclusions.
- Designed a reasoning-aware test-time decoding scheduler that improved GSM8K accuracy on LLaDA from **58.0%** to **76.4%** without updating the backbone model.

PUBLICATIONS

- **Zhenghao He**, Guangzhi Xiong, Sanchit Sinha, Bohan Liu, Wenqian Ye, Aidong Zhang. “Rethinking Reasoning Paths as Phase-Structured Trajectories.” Preprint, 2026.
- **Zhenghao He**, Guangzhi Xiong, Boyang Liu, Sanchit Sinha, Aidong Zhang. “CASL: Concept-Aligned Sparse Latents for Interpreting Diffusion Models.” Preprint, 2026.
- **Zhenghao He**, Guangzhi Xiong, Bohan Liu, Sanchit Sinha, Aidong Zhang. “Reasoning Beyond Chain-of-Thought: A Latent Computational Mode in Large Language Models.”The 2026 Conference on Empirical Methods in Natural Language Processing (**EMNLP Main**, 2026).
- **Zhenghao He**, Sanchit Sinha, Guangzhi Xiong, and Aidong Zhang. “GCAV: A Global Concept Activation Vector Framework for Cross-Layer Consistency in Interpretability.” *Proceedings of the IEEE/CVF International Conference on Computer Vision* ***(ICCV)**, 2025*, pp. 614-623.
- Bohan Liu, Wenqian Ye, Guangzhi Xiong, **Zhenghao He**, Sanchit Sinha, Aidong Zhang. “Towards Robustness against Typographic Attack with Training-free Concept Localization” , European Conference on Computer Vision (**ECCV**), 2026.
- Guangzhi Xiong, Sanchit Sinha, **Zhenghao He**, Aidong Zhang. “Retrieving Counterfactuals Improves Visual In-Context Learning. ” The IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**), 2026.
- Guangzhi Xiong, **Zhenghao He**, Bohan Liu, Sanchit Sinha, Aidong Zhang. “Toward Faithful Retrieval-Augmented Generation with Sparse Autoencoders.” International Conference on Learning Representations (***ICLR***), 2026.
- Sanchit Sinha, Guangzhi Xiong, **Zhenghao He**, and Aidong Zhang. “Concept-RuleNet: Grounded Multi-Agent Neurosymbolic Reasoning in Vision Language Models.” Proceedings of the AAAI Conference on Artificial Intelligence (***AAAI***), 2026. **Oral Presentation**.

- **Zhenghao He**, Ruifan Chen, Yayue Hou, Fei Xie, Xiaoliang Gong, and Anthony G. Cohn. "A Music Labeling Model Based on Traditional Chinese Music Characteristics for Emotional Regulation." *ICSCA*, 2023.
- Fei Xie, Xiaoliang Gong, **Zhenghao He**, Tongqi Wu, Yan Lu, and Mohan Zhao. "Alzheimer's Disease Early Diagnosis Based on Resting-State Dynamic Functional Connectivity." *ICSCA*, 2023.
- Mohan Zhao, Lu Pang, Yan Lu, Fei Xie, **Zhenghao He**, Xiaoliang Gong, and Anthony George Cohn. "Conditional Domain Adaptation Based on Initial Distribution Discrepancy for EEG Emotion Recognition." *MICCAI workshop.*
- **Zhenghao He**. "Comparison Of Different Machine Learning Methods Applied To Obesity Classification," *MLISE*, 2022.

WORK EXPERIENCE

Alibaba Cloud **Shanghai, China**
AI Model Optimization Intern *Mar 2024 – Aug 2024*

- Conducted a technical survey on GPU reliability and failure prediction methods in large-scale distributed clusters, reviewing monitoring approaches and machine learning techniques for anomaly detection.
- Analyzed cluster monitoring and telemetry data, including utilization, temperature, and memory metrics, to understand common operational patterns and potential indicators of GPU failures.
- Investigated AI agent architectures for adaptive task scheduling and multi-model orchestration, summarizing recent research and industry practices in cloud-based AI platforms.

Digital Center, HaaenClean Group **Shanghai, China**
Digital Engineer Intern *Sep 2023 – Dec 2023*

- Built and deployed an anomaly detection pipeline for multi-sensor industrial data.
- Developed an on-premise LLM-based system for secure document retrieval and automated report generation.

Shanghai Research Institute for Intelligent Autonomous Systems **Shanghai, China**
Research Assistant for Assoc Prof. Zhongpan Zhu, Tongji University *Jun 2022 – Sep 2022*

- Supported the development of research materials and technical presentations on bionic robotics.
- Proposed a patent "Algorithm of Generating Heat Map for Driver's Attention", focusing on computer vision-based driver monitoring systems.

HONORS & AWARDS

3rd Prize (School-level), 2023 Excellent Student Scholarship at Tongji University Dec 2023
2nd Prize (National Level), 2022 Embedded System Design Invitational Contest (Intel Cup) *Aug 2022*
Bronze Medal (University-level), China College Students Internet+ Innovation and Entrepreneurship Competition *Aug 2022*

OTHERS

Technical Skills: Python, PyTorch, TensorFlow, CUDA, C/C++, MATLAB; Git, Linux, LaTeX
Languages: Chinese (Native), English (Advanced)